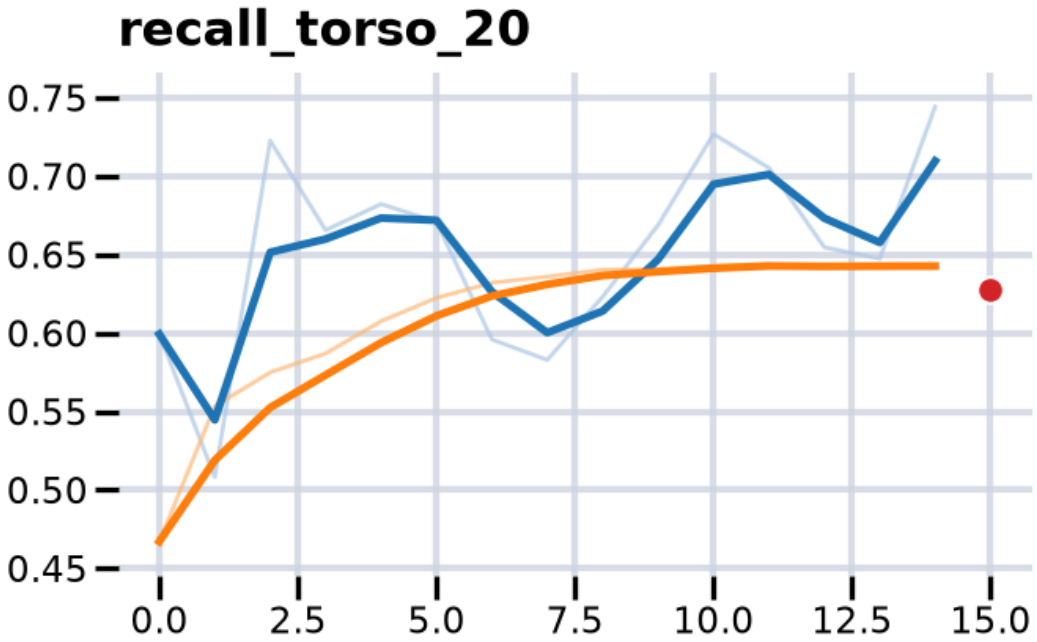
TMP04

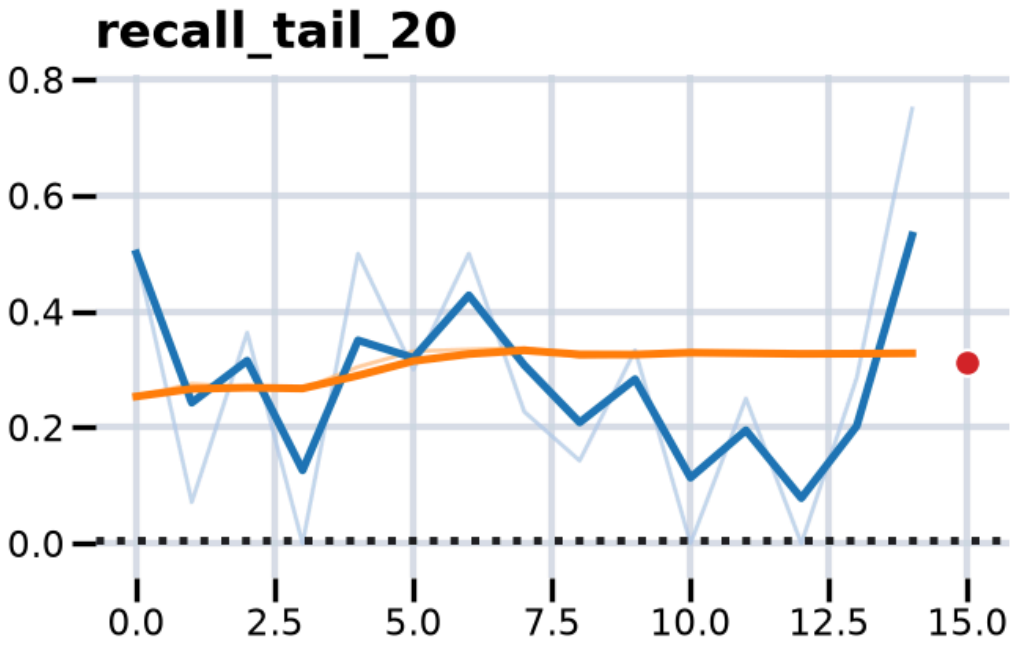
Best fitting hyper-params:

```
python3 -c "import json;
print(json.dumps(json.load(open('./hpo-results-bucket/kaggle-tune-train-test/kaggle-hpo/tune/hp
arams.json', 'r')), indent=4))"
{
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    "USER": "runner",
    "batch_size": 1024,
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    "debug": false,
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    "edge_embed_dim": 16,
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    "embed_len": 16,
    "focal_loss_gamma": 1.25,
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    "hpo_tier": 1,
    "latest_checkpoint_uri": null,
    "learning_rate": 0.0008,
    "max_history": 100,
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    "mlflow_experiment_name": "kaggle-hpo",
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    "mlp_hidden_dim": 1.5,
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    "movie_tiers_uri": "/tmp/gcs_data/data/movie_tiers-00000-of-00001.array_record",
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    "out_dim": 24,
    "output_hyperparams_uri": null,
    "output_metrics_uri": null,
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    "project_id": "kaggle-tune-train-test",
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    "ratings_test_disliked_uri": "/tmp/gcs_data/data/ratings_test_disliked.array_record",
    "ratings_test_liked_uri": "/tmp/gcs_data/data/ratings_test_liked.array_record",
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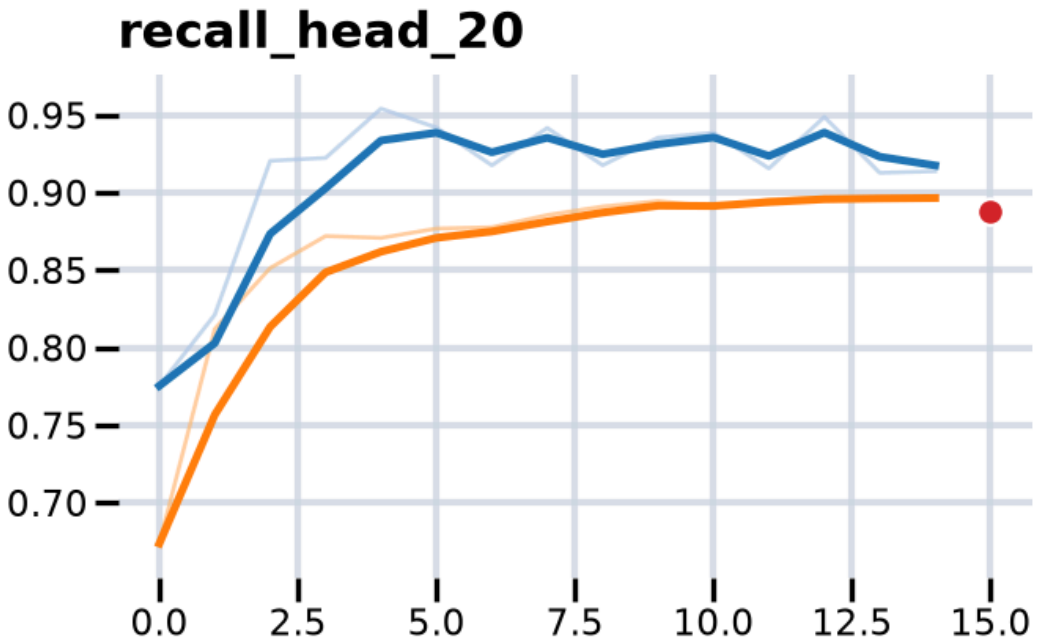
```
  "ratings_train_disliked_uri": "/tmp/gcs_data/data/ratings_train_disliked.array_record",
  "ratings_train_liked_uri": "/tmp/gcs_data/data/ratings_train_liked.array_record",
  "ratings_val_3_uri": "/tmp/gcs_data/data/ratings_val_3.array_record",
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  "temperature": 0.18,
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  "train_id": 0,
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  "trial_ids": [
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    1,
    2
  ],
  "use_focal_loss": 1,
  "use_ipw": 1,
  "user_embeddings_uri": "/tmp/gcs_data/data/user_emb-00000-of-00001.array_record",
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```



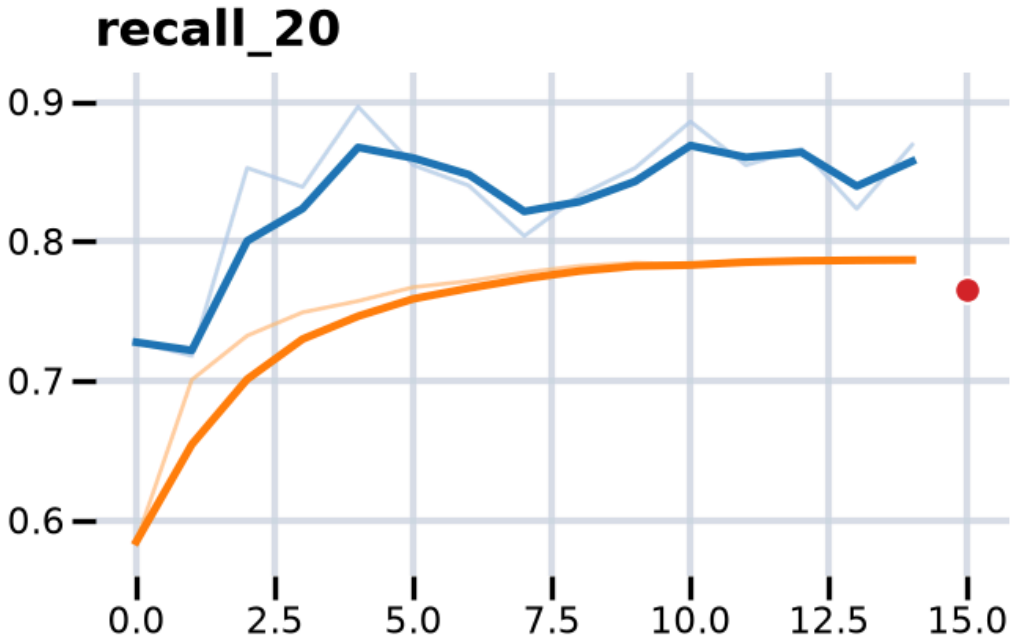

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.7097 | 0.7444 | 14 |
| validation | 0.6427 | 0.6427 | 14 |
| test | 0.6277 | 0.6277 | 15 |
| ceiling | - | - | - |
| random | - | 0.0052 | - |



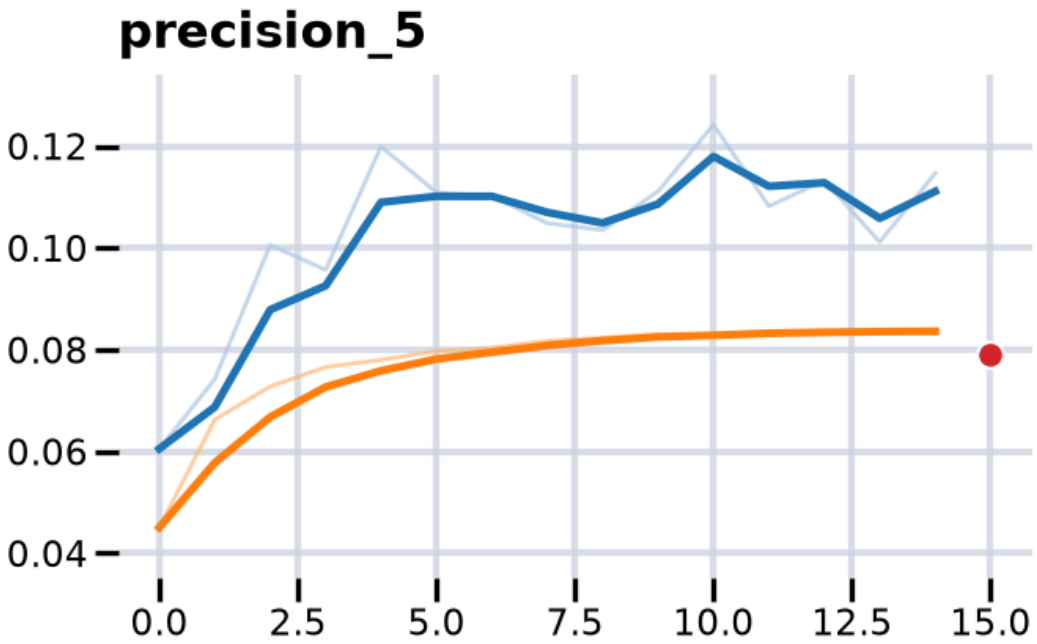

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.5311 | 0.7500 | 14 |
| validation | 0.3280 | 0.3284 | 14 |
| test | 0.3134 | 0.3134 | 15 |
| ceiling | - | - | - |
| random | - | 0.0052 | - |



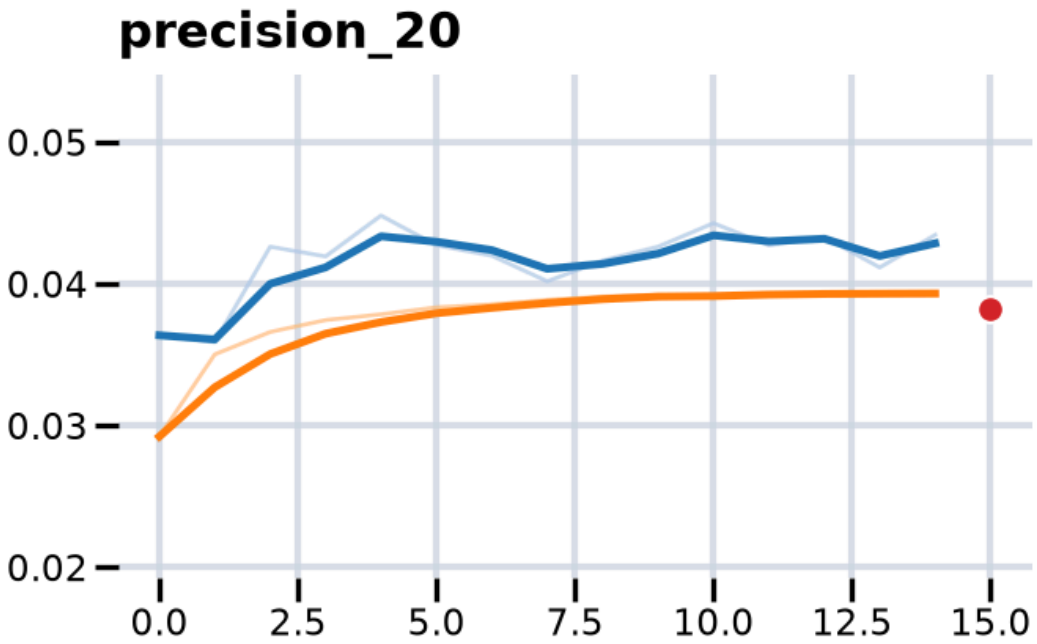

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.9175 | 0.9138 | 14 |
| validation | 0.8966 | 0.8967 | 14 |
| test | 0.8884 | 0.8884 | 15 |
| ceiling | - | - | - |
| random | - | 0.0052 | - |



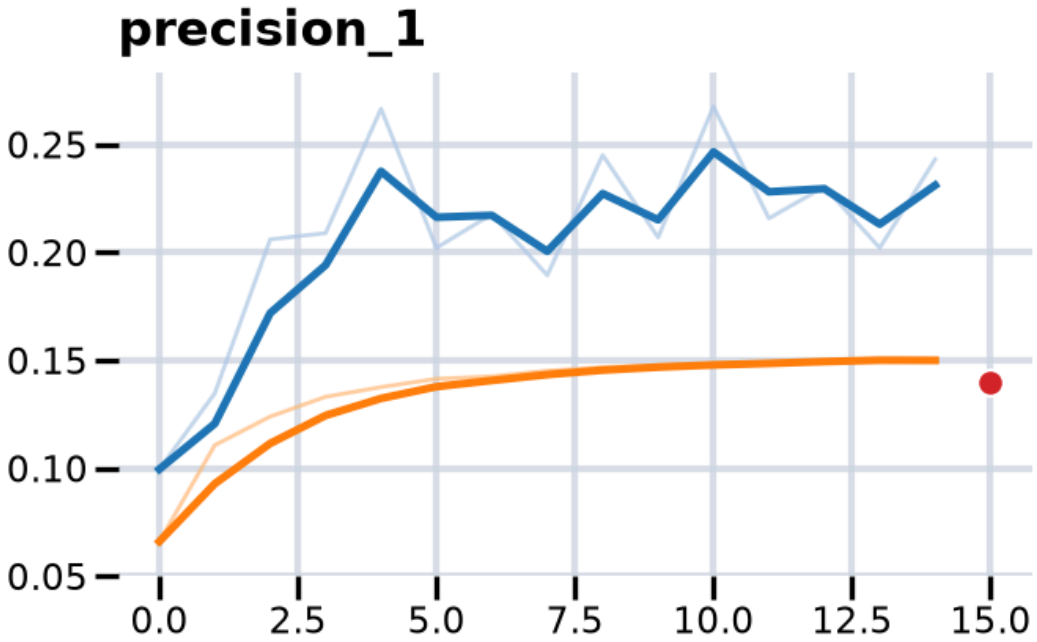

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.8573 | 0.8691 | 14 |
| validation | 0.7865 | 0.7866 | 14 |
| test | 0.7658 | 0.7658 | 15 |
| ceiling | - | - | - |
| random | - | 0.0052 | - |



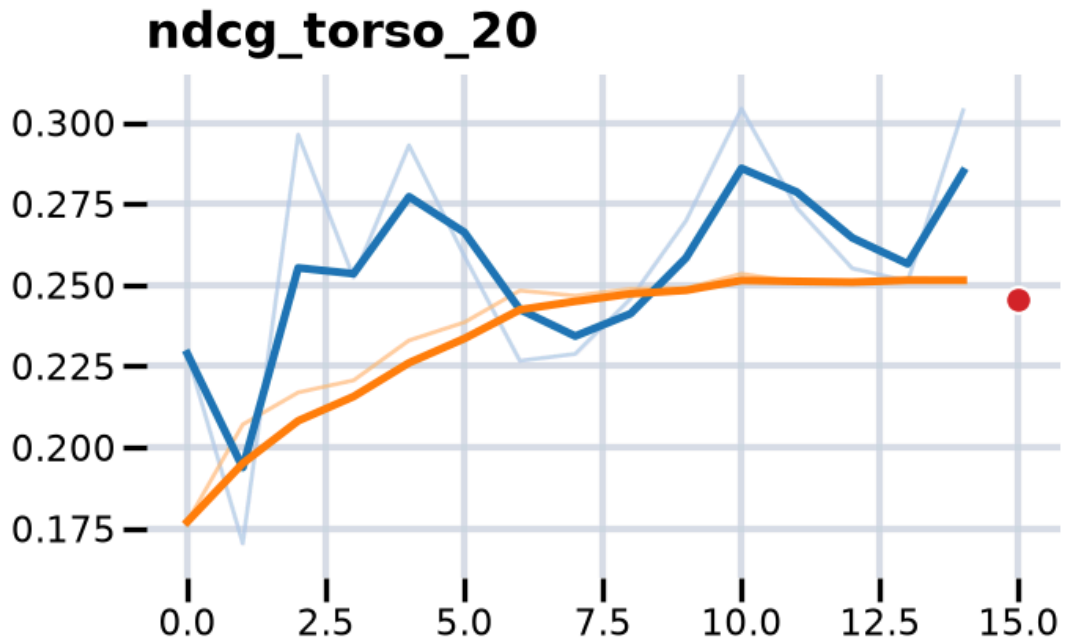

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.1111 | 0.1146 | 14 |
| validation | 0.0836 | 0.0837 | 14 |
| test | 0.0792 | 0.0792 | 15 |
| ceiling | - | - | - |
| random | - | 0.0003 | - |



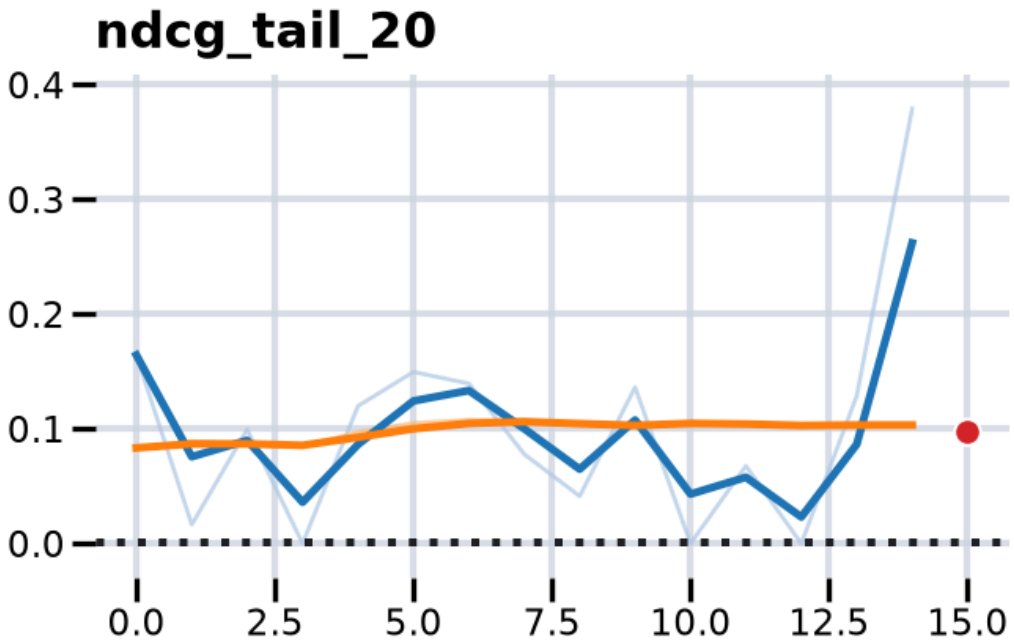

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.0429 | 0.0435 | 14 |
| validation | 0.0393 | 0.0393 | 14 |
| test | 0.0383 | 0.0383 | 15 |
| ceiling | - | - | - |
| random | - | 0.0003 | - |



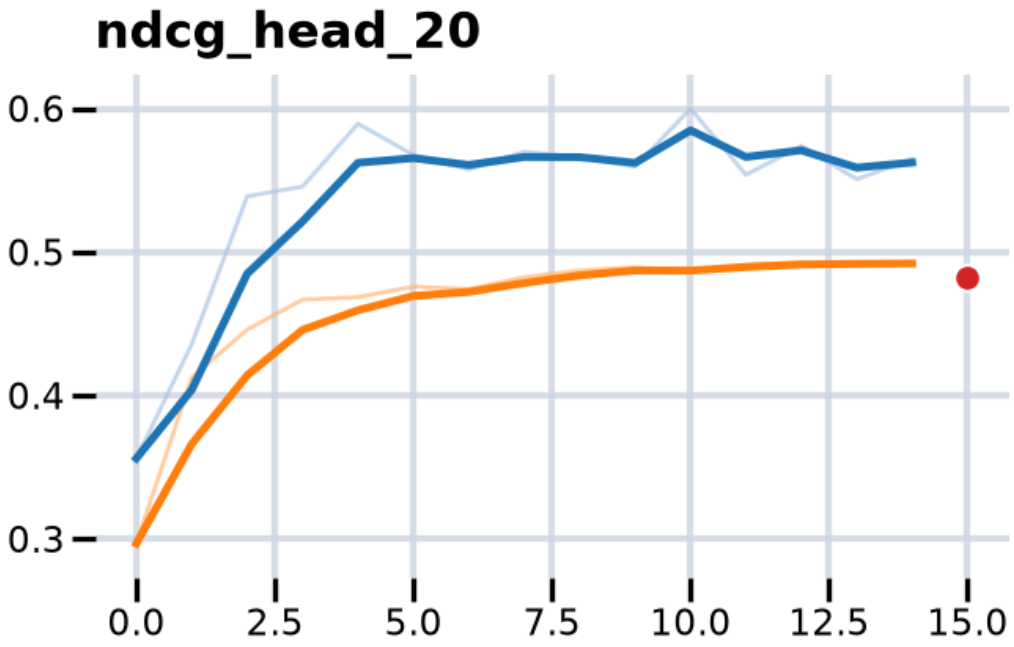

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.2311 | 0.2432 | 14 |
| validation | 0.1500 | 0.1500 | 14 |
| test | 0.1401 | 0.1401 | 15 |
| ceiling | - | - | - |
| random | - | 0.0003 | - |



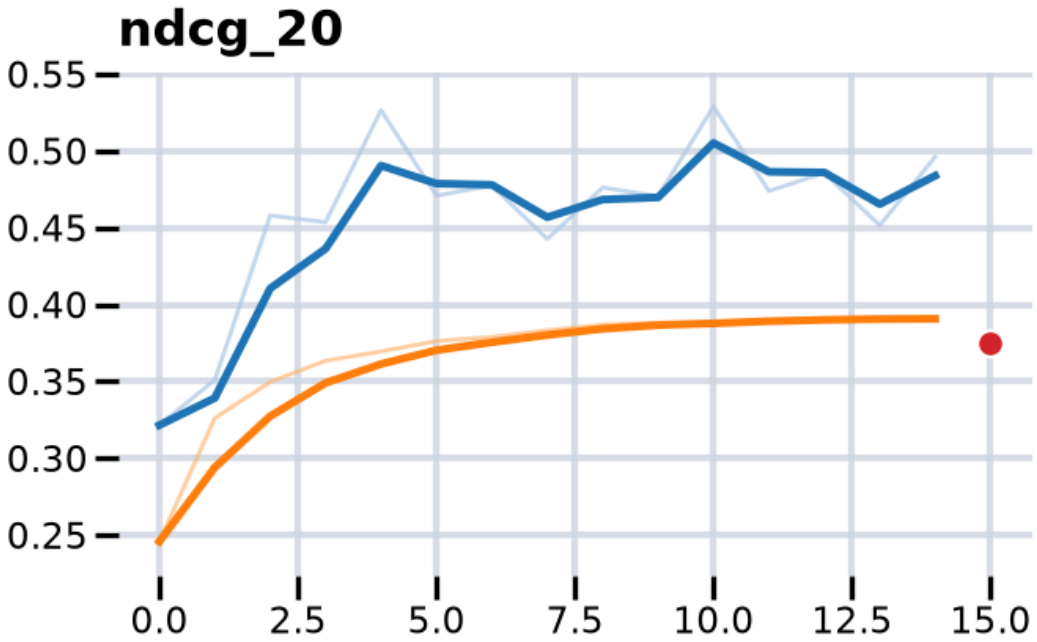

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.2849 | 0.3038 | 14 |
| validation | 0.2515 | 0.2515 | 14 |
| test | 0.2454 | 0.2454 | 15 |
| ceiling | - | - | - |
| random | - | 0.0018 | - |



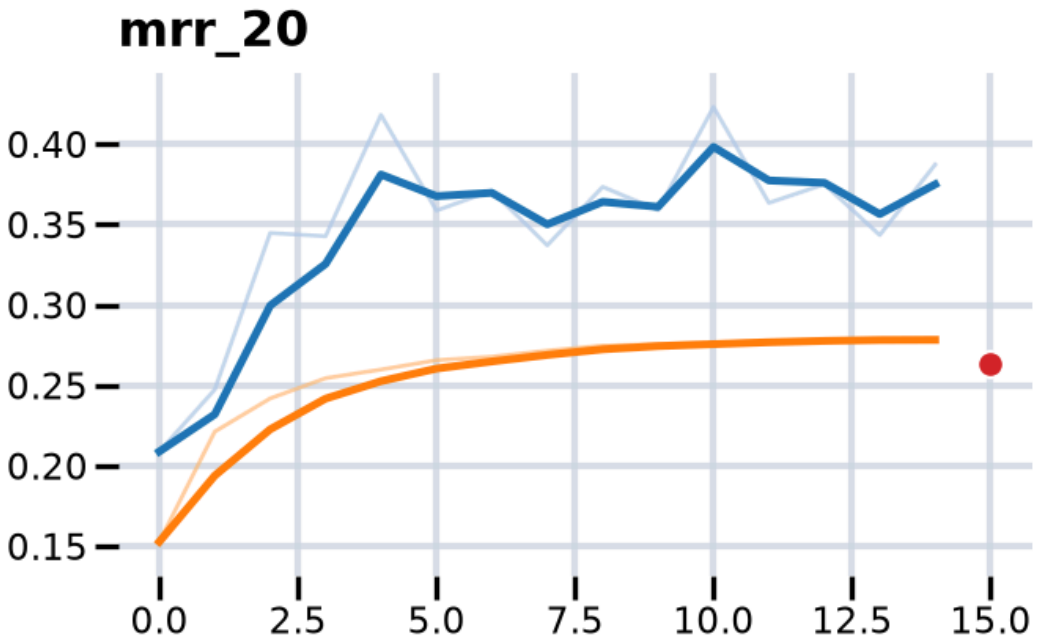

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.2621 | 0.3792 | 14 |
| validation | 0.1033 | 0.1033 | 14 |
| test | 0.0981 | 0.0981 | 15 |
| ceiling | - | - | - |
| random | - | 0.0018 | - |



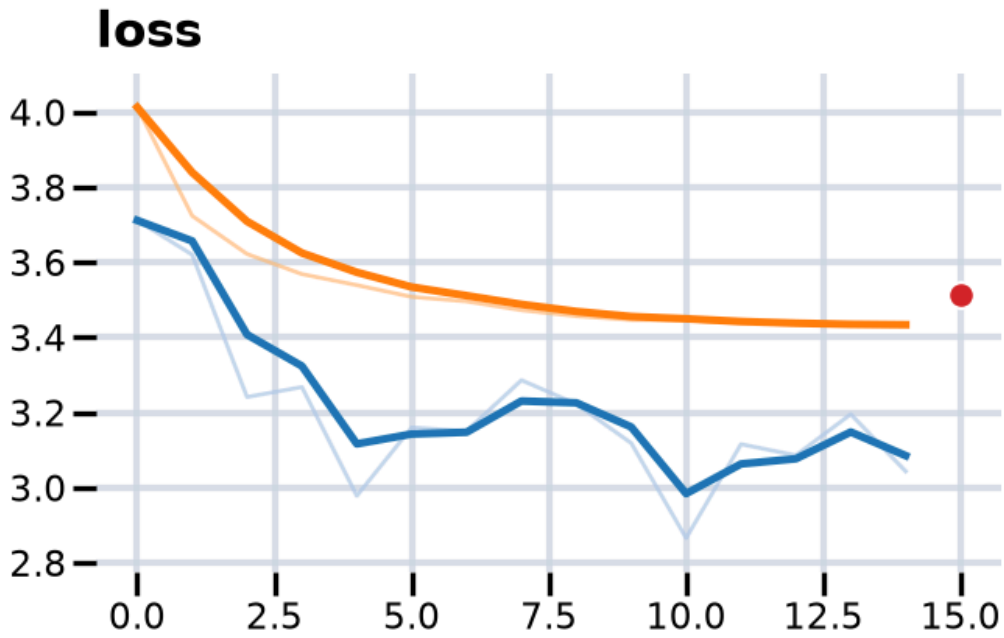

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.5623 | 0.5647 | 14 |
| validation | 0.4919 | 0.4921 | 14 |
| test | 0.4826 | 0.4826 | 15 |
| ceiling | - | - | - |
| random | - | 0.0018 | - |



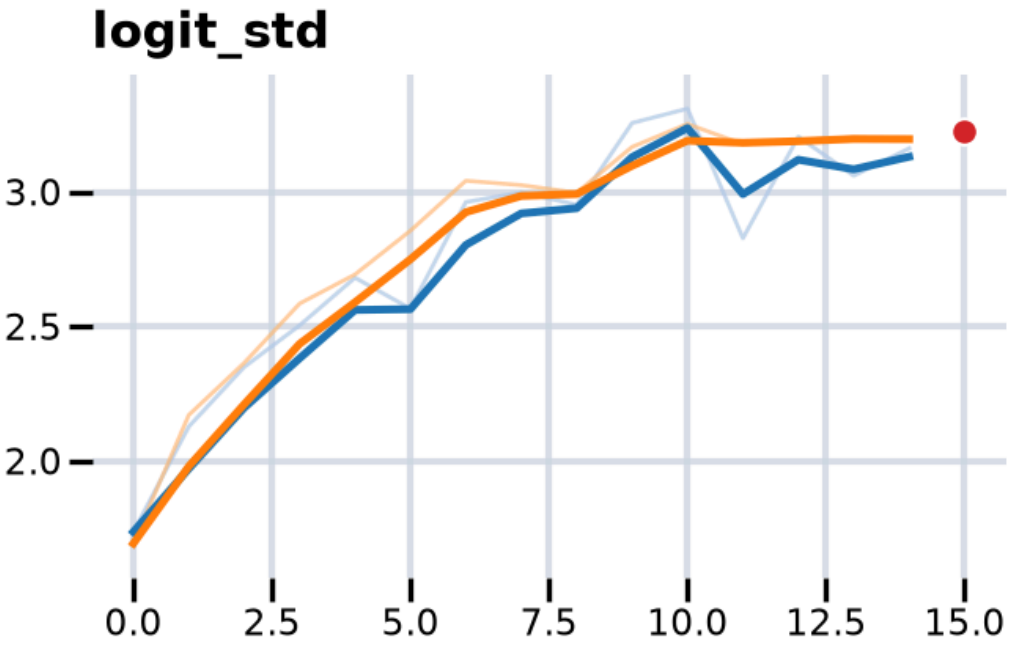

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.4838 | 0.4962 | 14 |
| validation | 0.3907 | 0.3908 | 14 |
| test | 0.3746 | 0.3746 | 15 |
| ceiling | - | - | - |
| random | - | 0.0018 | - |



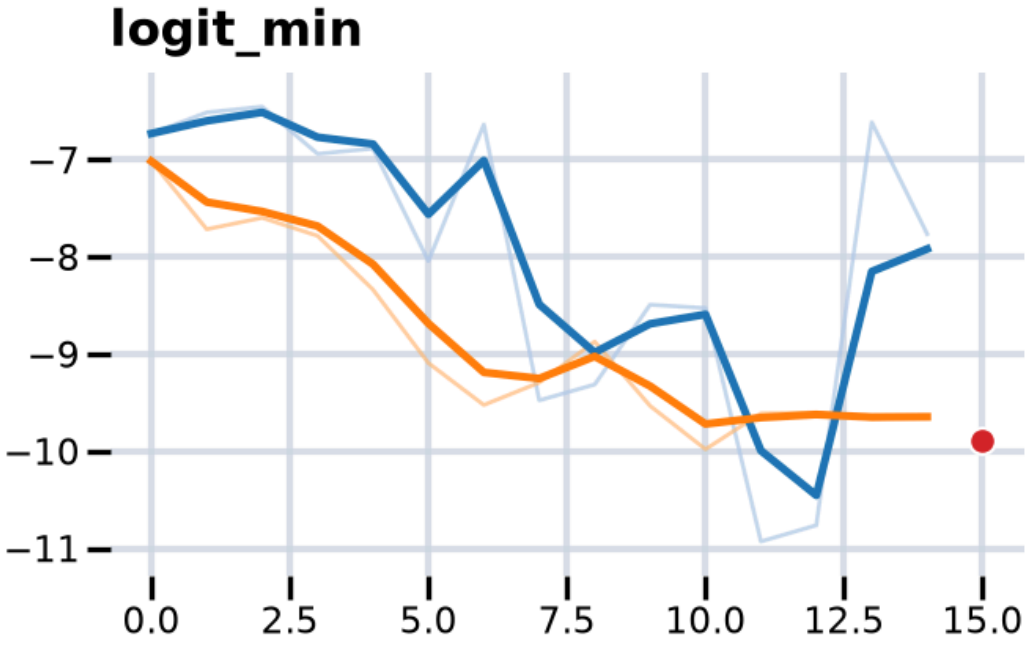

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.3750 | 0.3873 | 14 |
| validation | 0.2784 | 0.2784 | 14 |
| test | 0.2639 | 0.2639 | 15 |
| ceiling | - | - | - |
| random | - | 0.0009 | - |



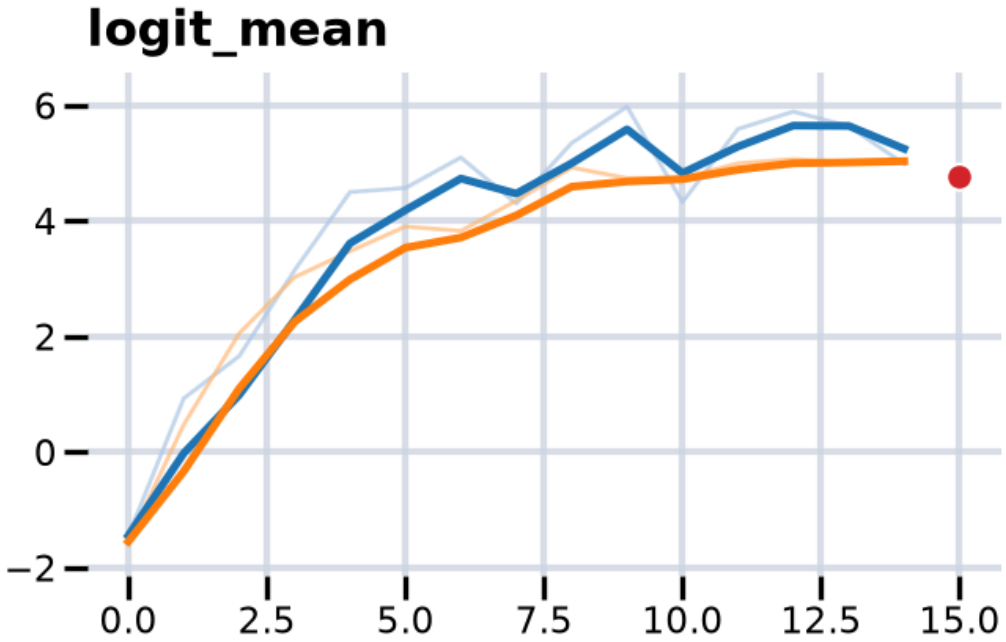

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 3.0856 | 3.0437 | 14 |
| validation | 3.4342 | 3.4335 | 14 |
| test | 3.5148 | 3.5148 | 15 |
| irred error | - | - | - |
| random | - | 4.6052 | - |



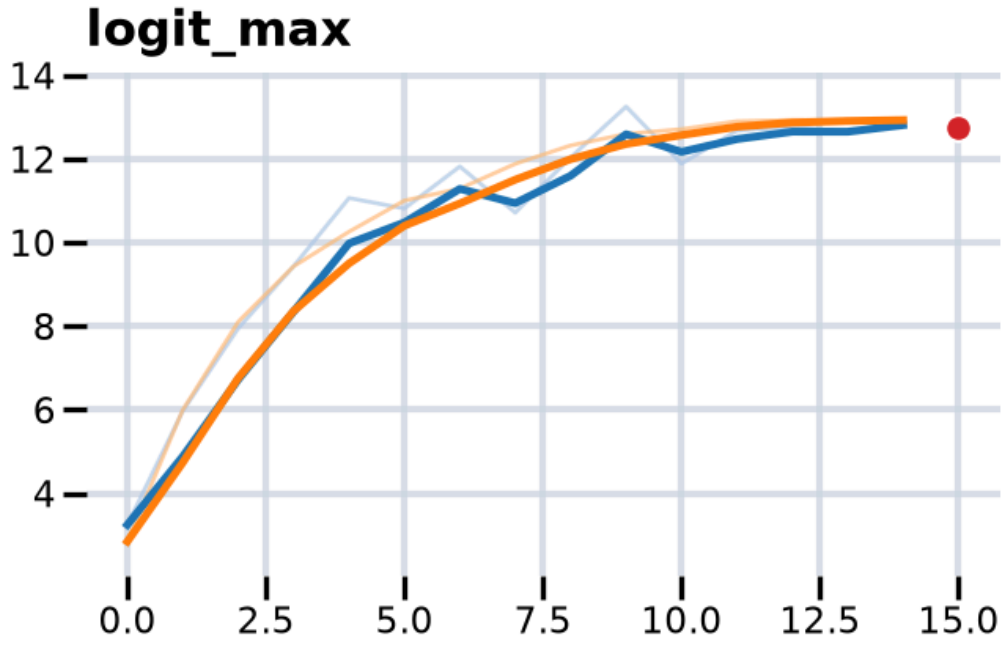

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 3.1302 | 3.1613 | 14 |
| validation | 3.1959 | 3.1955 | 14 |
| test | 3.2251 | 3.2251 | 15 |
| ceiling | - | - | - |



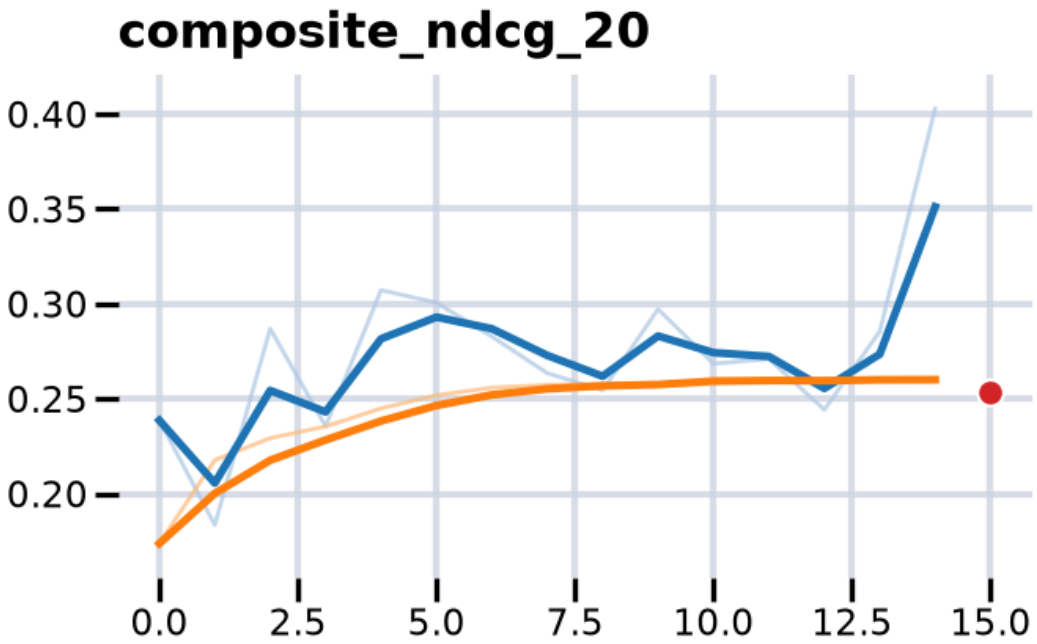

| | Smoothed | Value | Step |
|---|---|---|---|
| train | -7.9183 | -7.7637 | 14 |
| validation | -9.6459 | -9.6438 | 14 |
| test | -9.8942 | -9.8942 | 15 |
| ceiling | - | - | - |



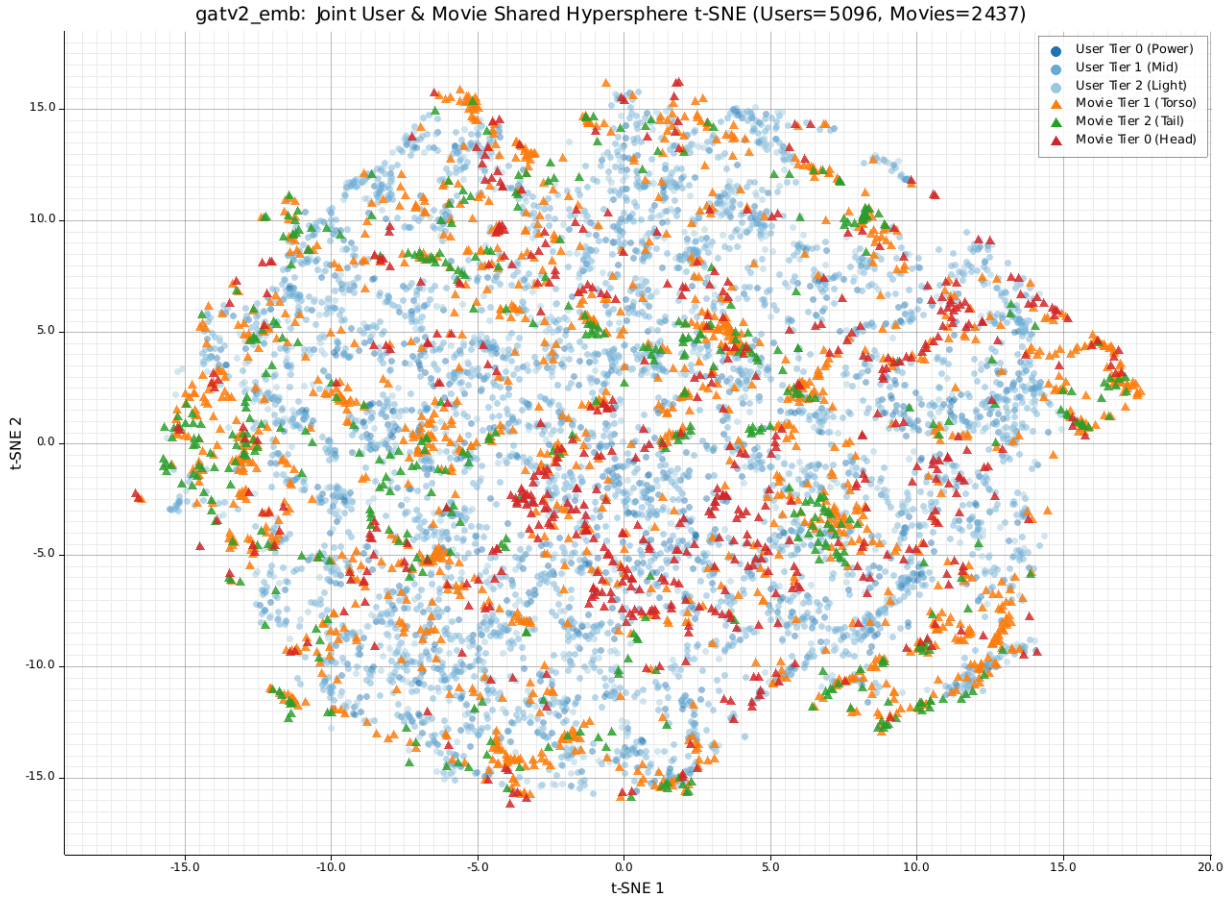

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 5.2502 | 4.9866 | 14 |
| validation | 5.0356 | 5.0500 | 14 |
| test | 4.7739 | 4.7739 | 15 |
| ceiling | - | - | - |

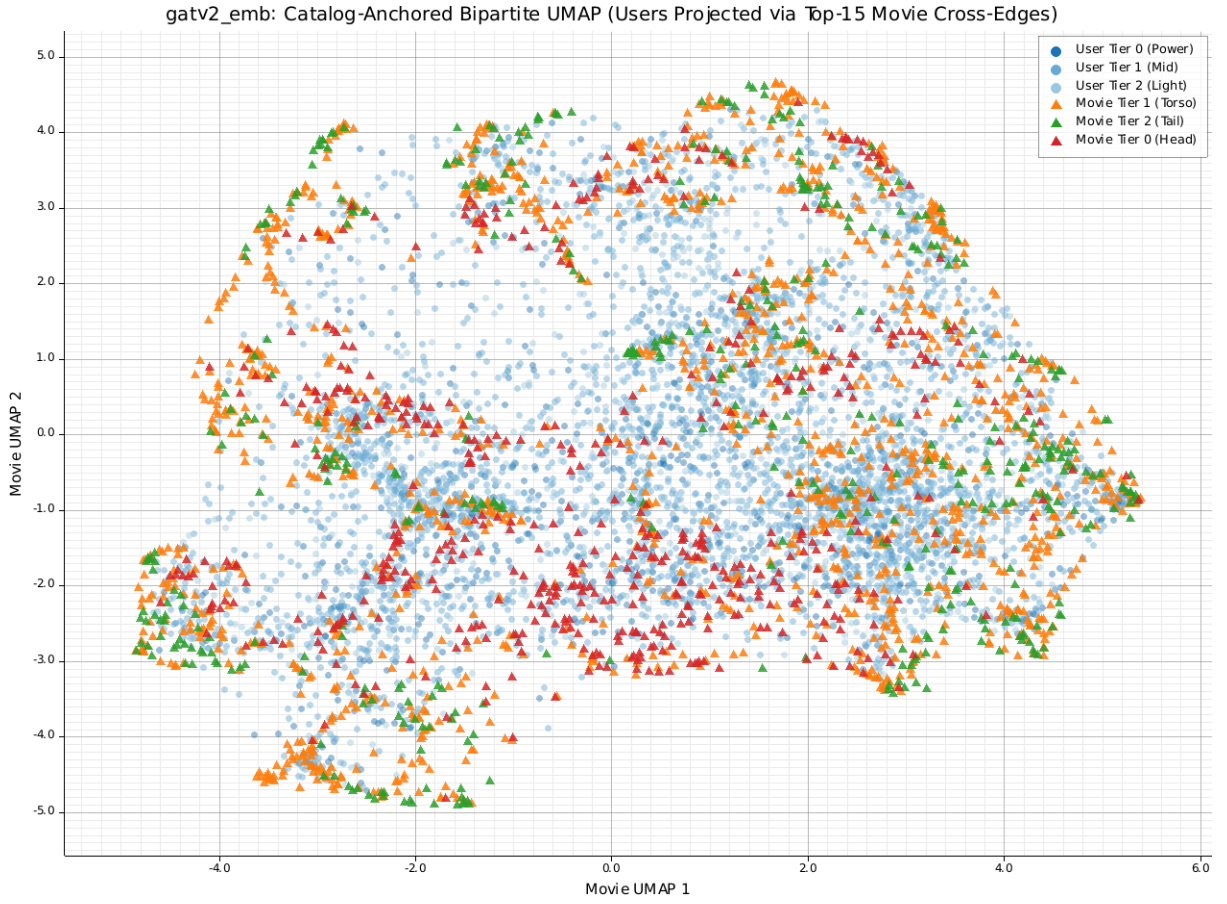


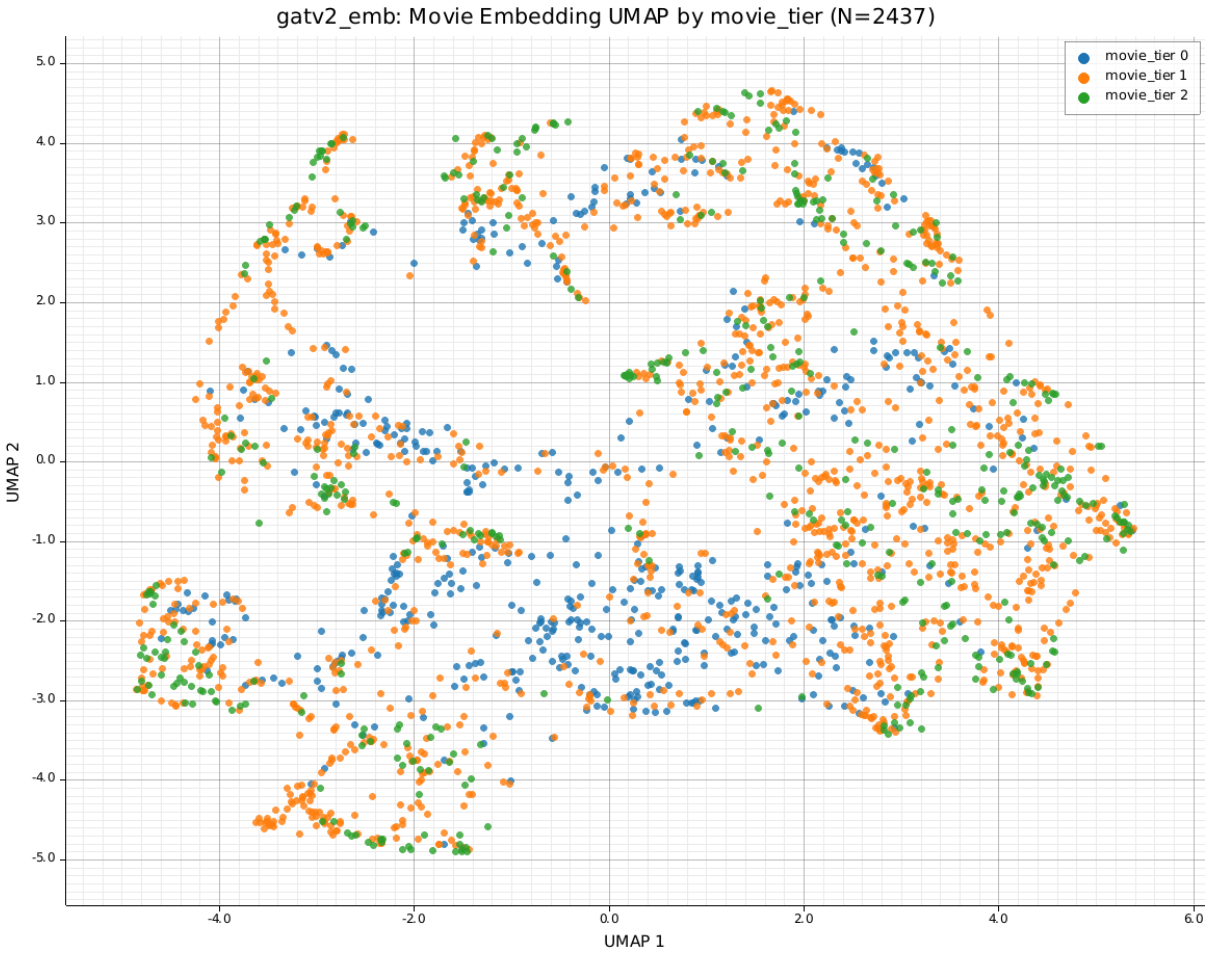

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 12.8116 | 12.9180 | 14 |
| validation | 12.9308 | 12.9452 | 14 |
| test | 12.7420 | 12.7420 | 15 |
| ceiling | - | - | - |

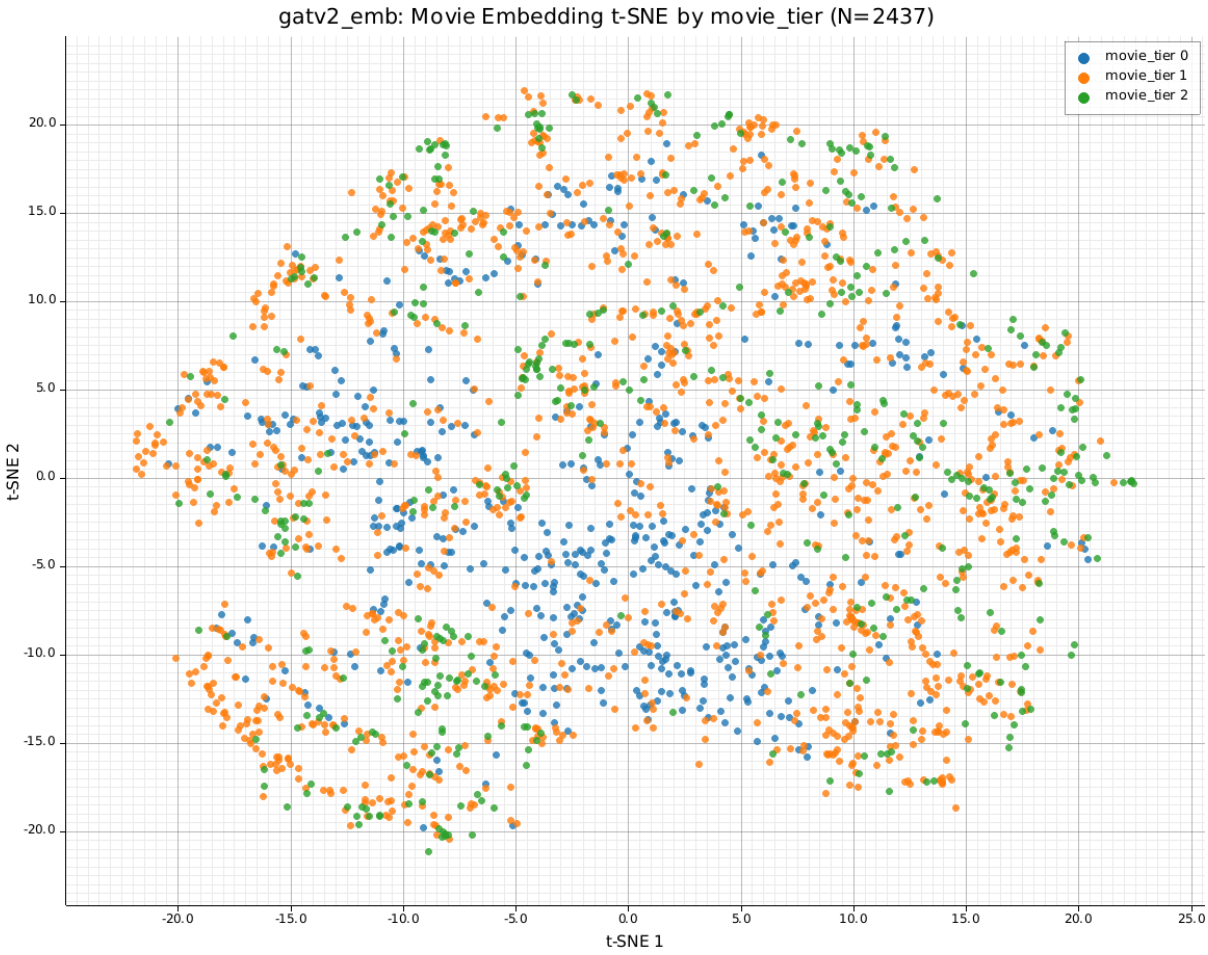


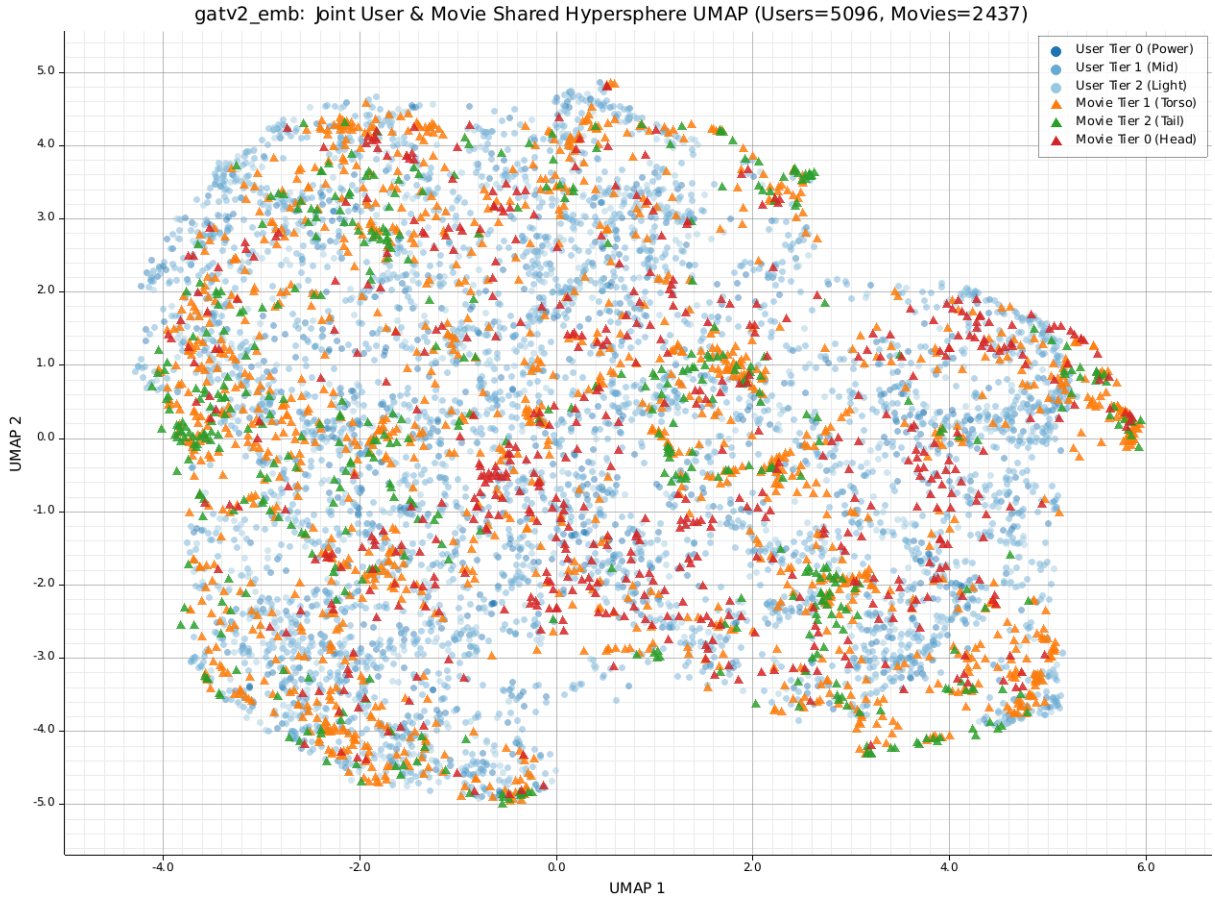

| | Smoothed | Value | Step |
|---|---|---|---|
| train | 0.3511 | 0.4029 | 14 |
| validation | 0.2601 | 0.2601 | 14 |
| test | 0.2535 | 0.2535 | 15 |
| ceiling | - | - | - |
| random | - | 0.0018 | - |

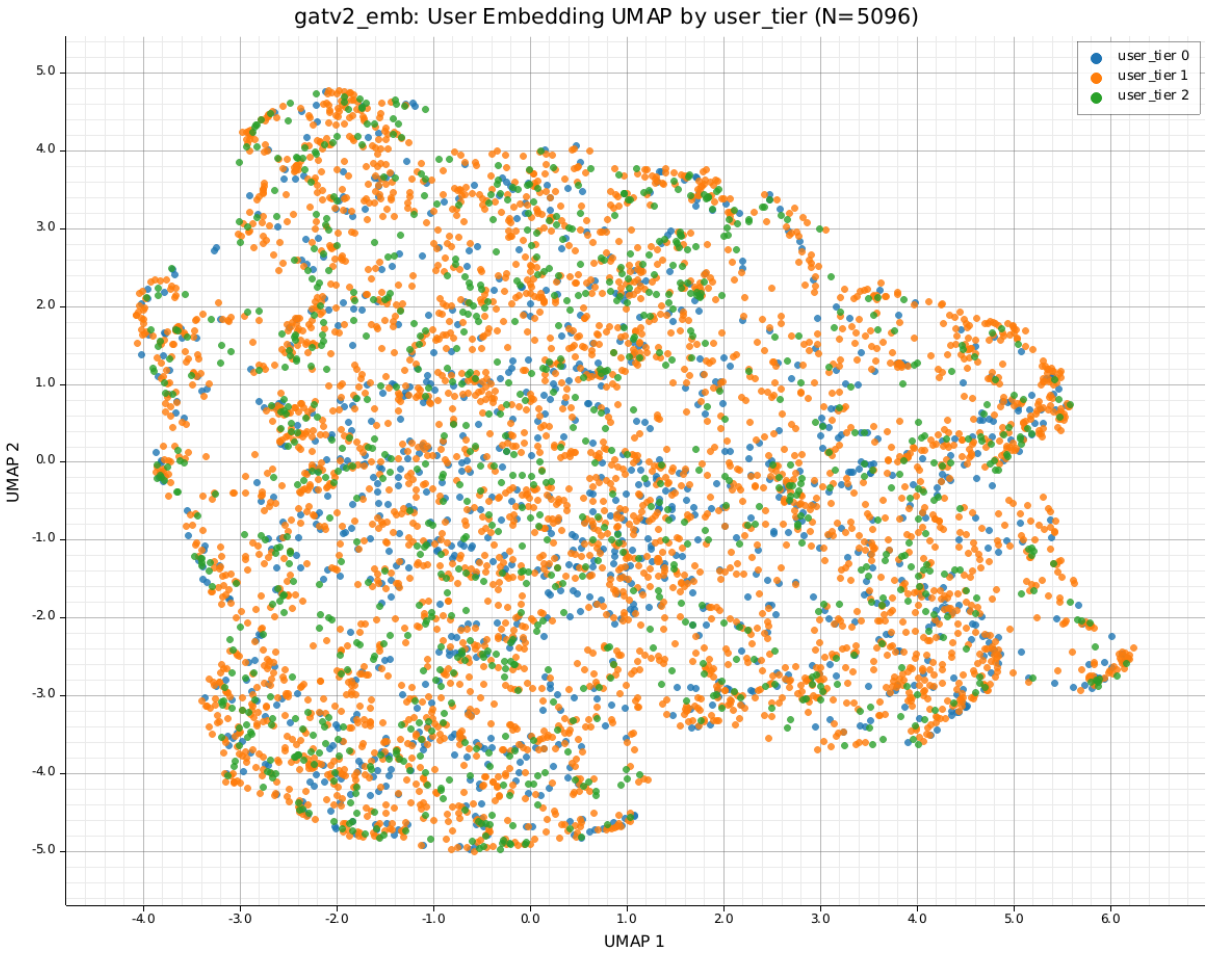

gatv2_emb: Joint User & Movie Shared Hypersphere t-SNE (Users=5096, Movies=2437)
User Tier 0 (Power)
User Tier 1 (Mid)
User Tier 2 (Light)
Movie Tier 1 (Torso)
Movie Tier 2 (Tail)
Movie Tier 0 (Head)
t-SNE 2
t-SNE 1
15.0
10.0
5.0
0.0
-5.0
-10.0
-15.0
-15.0
-10.0
-5.0
0.0
5.0
10.0
15.0
20.0

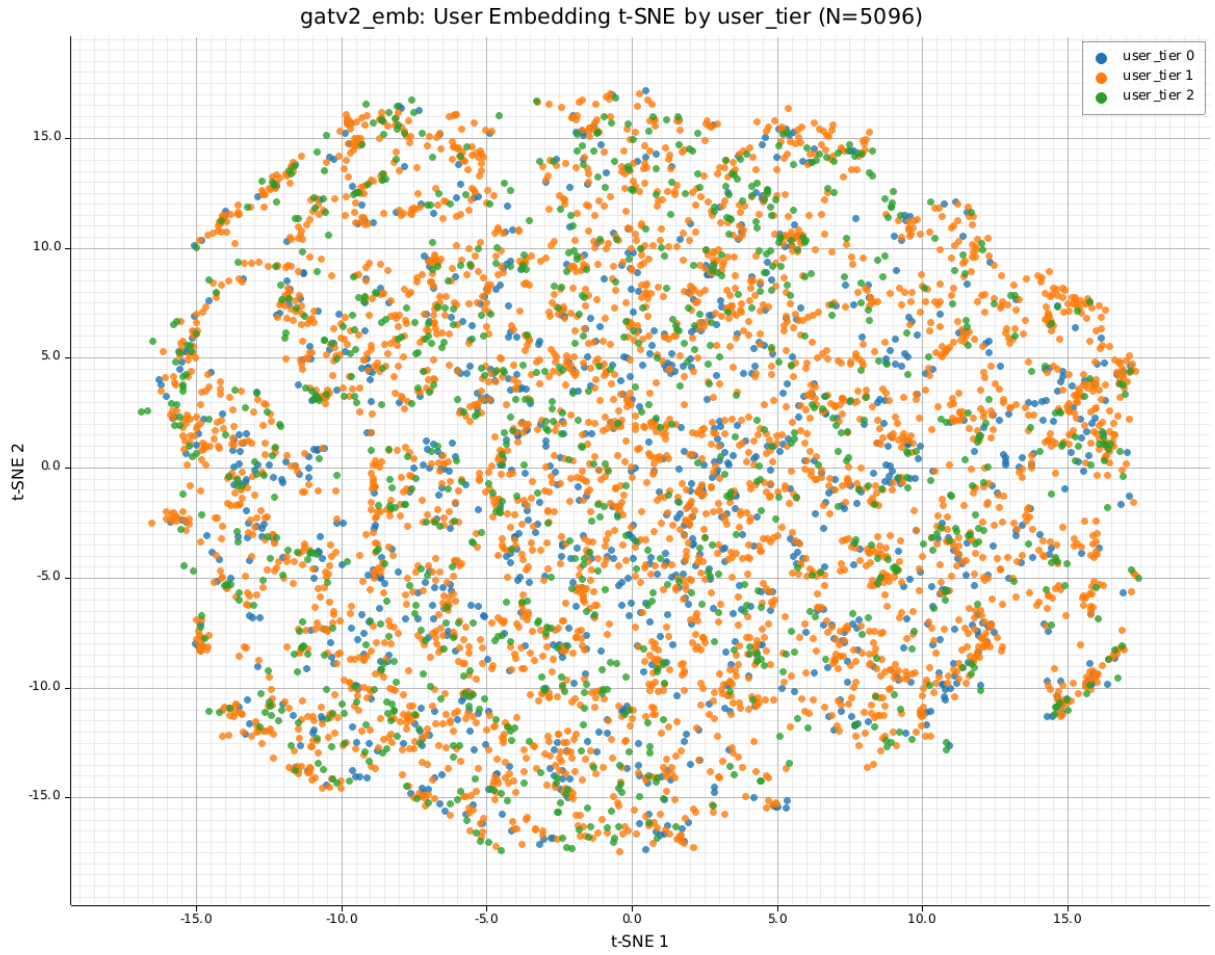

gatv2_emb: Catalog-Anchored Bipartite UMAP (Users Projected via Top-15 Movie Cross-Edges)
User Tier 0 (Power)
User Tier 1 (Mid)
User Tier 2 (Light)
Movie Tier 1 (Torso)
Movie Tier 2 (Tail)
Movie Tier 0 (Head)
Movie UMAP 2
Movie UMAP 1
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
-4.0
-2.0
0.0
2.0
4.0
6.0


gatv2_emb: Movie Embedding UMAP by movie_tier (N=2437)
movie_tier 0
movie_tier 1
movie_tier 2
UMAP 2
UMAP 1
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
-4.0
-2.0
0.0
2.0
4.0
6.0


gatv2_emb: Movie Embedding t-SNE by movie_tier (N=2437)
movie_tier 0
movie_tier 1
movie_tier 2
t-SNE 2
t-SNE 1
20.0
15.0
10.0
5.0
0.0
-5.0
-10.0
-15.0
-20.0
-20.0
-15.0
-10.0
-5.0
0.0
5.0
10.0
15.0
20.0
25.0


gatv2_emb: Joint User & Movie Shared Hypersphere UMAP (Users=5096, Movies=2437)
User Tier 0 (Power)
User Tier 1 (Mid)
User Tier 2 (Light)
Movie Tier 1 (Torso)
Movie Tier 2 (Tail)
Movie Tier 0 (Head)
UMAP 2
UMAP 1
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
-4.0
-2.0
0.0
2.0
4.0
6.0


gatv2_emb: User Embedding UMAP by user_tier (N=5096)
user_tier 0
user_tier 1
user_tier 2
UMAP 2
UMAP 1
5.0
4.0
3.0
2.0
1.0
0.0
-1.0
-2.0
-3.0
-4.0
-5.0
-4.0
-3.0
-2.0
-1.0
0.0
1.0
2.0
3.0
4.0
5.0
6.0





{
  "cold_start_movies": {
    "ranker_20_cold_start_movies": [
      "STRONG RECALL LIFT: Global Recall@20 (3.7%) heavily outperforms the random expected baseline (0.5%). The candidate generator is extracting meaningful semantic signal.",
      "STRONG PRECISION LIFT: Global Precision@20 (1.65%) heavily outperforms the random expected precision (0.22%), indicating high density of relevant items in the retrieved slates.",
      "SLOW SURFACING WARNING: Global MRR@20 is heavily degraded (0.07), indicating users must scroll significantly to find the first relevant item.",
      "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 4.6%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 3.5%).",
      "HEALTHY RANKING STRATIFICATION: NDCG is higher for power users (Tier 0: 4.94% vs Tier 2: 2.35%). The model not only retrieves the right items for power users but places them higher in the slate.",
      "POPULARITY BIAS DETECTED: Ranking quality heavily favors head/popular movies (Tier 0 NDCG: 2.69%) compared to long-tail/niche items (Tier 2 NDCG: 0.31%)."
    ],
    "retrieval_ann_100_cold_start_movies": [
      "STRONG RECALL LIFT: Global Recall@100 (8.7%) heavily outperforms the random expected baseline (2.3%). The candidate generator is extracting meaningful semantic signal.",
      "STRONG PRECISION LIFT: Global Precision@100 (0.75%) heavily outperforms the random expected precision (0.22%), indicating high density of relevant items in the retrieved slates.",

```
    "SLOW SURFACING WARNING: Global MRR@100 is heavily degraded (0.04), indicating users must scroll significantly to find the first relevant item.",
    "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 9.9%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 7.9%).",
    "HEALTHY RANKING STRATIFICATION: NDCG is higher for power users (Tier 0: 5.04% vs Tier 2: 2.93%). The model not only retrieves the right items for power users but places them higher in the slate.",
    "BALANCED ITEM TIER RANKING: NDCG performance remains consistent across movie tiers (Head Tier 0: 2.48%, Tail Tier 2: 1.84%), indicating stable retrieval across popular and long-tail catalogs."
   ],
   "retrieval_ann_20_cold_start_movies": [
    "STRONG RECALL LIFT: Global Recall@20 (2.4%) heavily outperforms the random expected baseline (0.5%). The candidate generator is extracting meaningful semantic signal.",
    "STRONG PRECISION LIFT: Global Precision@20 (1.07%) heavily outperforms the random expected precision (0.22%), indicating high density of relevant items in the retrieved slates.",
    "SLOW SURFACING WARNING: Global MRR@20 is heavily degraded (0.04), indicating users must scroll significantly to find the first relevant item.",
    "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 2.5%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 2.3%).",
    "HEALTHY RANKING STRATIFICATION: NDCG is higher for power users (Tier 0: 2.79% vs Tier 2: 1.24%). The model not only retrieves the right items for power users but places them higher in the slate.",
    "BALANCED ITEM TIER RANKING: NDCG performance remains consistent across movie tiers (Head Tier 0: 1.12%, Tail Tier 2: 0.86%), indicating stable retrieval across popular and long-tail catalogs."
   ]
  },
  "cold_start_users": {
   "ranker_20_cold_start_users": [
    "MODERATE RECALL LIFT: Global Recall@20 (1.5%) beats the random baseline (0.5%), but there is room for improvement.",
    "MODERATE PRECISION LIFT: Global Precision@20 (1.03%) is better than random (0.39%), suggesting basic relevance filtering is functioning.",
    "SLOW SURFACING WARNING: Global MRR@20 is heavily degraded (0.04), indicating users must scroll significantly to find the first relevant item.",
    "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 2.9%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 0.8%).",
    "STRONG NICHE USER RANKING: Light/niche users achieve strong top-K ranking quality (Tier 2 NDCG: 3.00% vs Tier 0: 1.41%).",
    "POPULARITY BIAS DETECTED: Ranking quality heavily favors head/popular movies (Tier 0 NDCG: 1.65%) compared to long-tail/niche items (Tier 2 NDCG: 0.00%)."
   ],
   "retrieval_ann_100_cold_start_users": [
    "RECALL FAILURE: The model fails to outperform a random candidate generator on Recall.",
    "PRECISION FAILURE: The model retrieves slates with the same or worse precision than a completely random draw.",
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    "SLOW SURFACING WARNING: Global MRR@100 is heavily degraded (0.01), indicating users must scroll significantly to find the first relevant item.",
    "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 4.3%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 1.8%).",
    "STRONG NICHE USER RANKING: Light/niche users achieve strong top-K ranking quality (Tier 2 NDCG: 2.84% vs Tier 0: 1.03%).",
    "POPULARITY BIAS DETECTED: Ranking quality heavily favors head/popular movies (Tier 0 NDCG: 0.95%) compared to long-tail/niche items (Tier 2 NDCG: 0.38%)."
   ],
   "retrieval_ann_20_cold_start_users": [
    "MODERATE RECALL LIFT: Global Recall@20 (0.6%) beats the random baseline (0.5%), but there is room for improvement.",
    "PRECISION FAILURE: The model retrieves slates with the same or worse precision than a completely random draw.",
    "SLOW SURFACING WARNING: Global MRR@20 is heavily degraded (0.01), indicating users must scroll significantly to find the first relevant item.",
    "STRONG LIGHT-USER GENERALIZATION: Light/niche users (Tier 2 Recall: 2.9%) achieve recall comparable to or exceeding power users (Tier 0 Recall: 0.1%).",
    "STRONG NICHE USER RANKING: Light/niche users achieve strong top-K ranking quality (Tier 2 NDCG: 2.47% vs Tier 0: 0.21%).",
    "POPULARITY BIAS DETECTED: Ranking quality heavily favors head/popular movies (Tier 0 NDCG: 0.30%) compared to long-tail/niche items (Tier 2 NDCG: 0.00%)."
   ]
  },
  "contextual_hubness": [
   "HEALTHY EXPRESSIVITY (ranker_contextual_k_20): The mean contextual similarity is 0.45. The GATv2 layers are successfully preserving geometric breadth between users and candidates."
  ],
  "coverage_and_gini": {
   "ranker_20": [
    "HEALTHY BIAS at_20 (ranker_20): Gini is 0.78. The model successfully balances mainstream relevance with catalog exploration.",
    "STRONG TAIL at_20 (ranker_20): The bottom 80% captures 17.7% of traffic, indicating excellent long-tail surfacing capability.",
    "DIVERSE HEAD at_20 (ranker_20): The top 10% consume 59.7% of slots, leaving plenty of room for the torso/tail.",
    "UNIFORM COHORTS at_20 (ranker_20): Light and Power users experience roughly the same level of catalog concentration."
   ],
   "retrieval_ann_100": [
    "HEALTHY BIAS at_100 (retrieval_ann_100): Gini is 0.48. The model successfully balances mainstream relevance with catalog exploration.",
    "STRONG TAIL at_100 (retrieval_ann_100): The bottom 80% captures 51.4% of traffic, indicating excellent long-tail surfacing capability.",
    "DIVERSE HEAD at_100 (retrieval_ann_100): The top 10% consume 29.9% of slots, leaving plenty of room for the torso/tail.",
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    "WARNING (COHORT COLLAPSE) at_100 (retrieval_ann_100): Power users have higher concentration (Gini) than light users. The model may be pulling rich histories into dense popularity traps."
   ],
   "retrieval_ann_20": [
    "HEALTHY BIAS at_20 (retrieval_ann_20): Gini is 0.63. The model successfully balances mainstream relevance with catalog exploration.",
    "STRONG TAIL at_20 (retrieval_ann_20): The bottom 80% captures 36.3% of traffic, indicating excellent long-tail surfacing capability.",
    "DIVERSE HEAD at_20 (retrieval_ann_20): The top 10% consume 43.5% of slots, leaving plenty of room for the torso/tail.",
    "UNIFORM COHORTS at_20 (retrieval_ann_20): Light and Power users experience roughly the same level of catalog concentration."
   ]
  },
  "cross_metric_analysis": [
   "FUNNEL RETENTION: Shrinking the candidate pool from 100 to 20 retains 17.2% of the recall (Recall drops from 1.60% to 0.27%).",
   "SURFACING SPEED LIFT: The Ranker improved MRR@20 by 396.3% (from 0.01 to 0.03). The Cross-Encoder successfully pushes the first relevant item much closer to rank 1.",
   "RANKER LIFT: The Ranker improved NDCG@20 by 377.3% over pure retrieval. The Cross-Encoder is successfully refining the candidate slates.",
   "SUCCESSFUL FILTERING: The Ranker aggressively filtered out 100.0% of the Retriever's items, resulting in a positive NDCG lift of 377.3%. It successfully separated the signal from the ANN's noise.",
   "RANKER COVERAGE COLLAPSE: To achieve its ranking, the Ranker sacrificed 29.6% of the Retriever's catalog coverage. It is behaving much more conservatively than the ANN.",
   "ITEM COLD-START RESILIENCE (RETRIEVER): When 501 test movies are replaced with unmemorized OOV=0 phantom items, Recall@100 retains 545.2% of warm performance (8.71% vs 1.60%). The Candidate Tower's content features (genres/metadata) generalize strongly into the embedding space.",
   "ITEM COLD-START RANKER LIFT: Even with OOV=0 phantom movies in the candidate pool, the Ranker improves NDCG@20 by 65.1% (and MRR@20 by 80.4%) over pure retrieval. The GATv2 graph attention effectively scores cold-start item embeddings against user history.",
   "HEAD ITEM ID DEPENDENCY: Under cold-start movie simulation, Tier 0 (head) NDCG@20 drops by -135.3% (1.14% -> 2.69%), compared to a -192.8% change for Tier 2 (long-tail) items. Popular items rely heavily on memorized collaborative ID weights rather than metadata.",
   "STRONG DEMOGRAPHIC COLD-START (USER RETRIEVER): With user_id=0 (OOV), demographic features alone (age, gender, occupation) retain 157.3% of warm Recall@100 (2.51% vs 1.60%, 1.0x random baseline).",
   "ZERO-HISTORY GRAPH RANKER LIFT: Even when cold-start users have an empty watch history in the GATv2 subgraph, the Ranker improves NDCG@20 by 307.2% (and MRR@20 by 523.9%) over pure retrieval.",
   "POWER USER HISTORY ABLATION: Stripping Power Users (Tier 0) of their interaction history and user ID reduces Ranker Recall@20 from 1.39% to 0.84% (confirming that learned history + ID provides a +64.4% relative lift over raw demographics).",
   "COLD-START BOTTLENECK (USER SIDE): End-to-end Ranker NDCG@20 retention is significantly lower for Cold-Start Users (156.8% of warm) than Cold-Start Movies (279.9% of
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warm). Onboarding new users (e.g., via interactive preference elicitation or richer user features) is a higher-leverage priority than item metadata."
  ],
  "diversity_metrics": {
    "ranker_20": [
      "HIGH PERSONALIZATION (ranker_20): Global Inter-list Diversity is 98.54%. The model successfully delivers highly unique slates customized to individual users.",
      "HIGH BREADTH (ranker_20): Intra-list Diversity is high (0.69), indicating slates feature a wide variety of semantic genres/topics."
    ],
    "retrieval_ann_100": [
      "HIGH PERSONALIZATION (retrieval_ann_100): Global Inter-list Diversity is 97.32%. The model successfully delivers highly unique slates customized to individual users.",
      "HIGH BREADTH (retrieval_ann_100): Intra-list Diversity is high (0.64), indicating slates feature a wide variety of semantic genres/topics."
    ],
    "retrieval_ann_20": [
      "HIGH PERSONALIZATION (retrieval_ann_20): Global Inter-list Diversity is 99.15%. The model successfully delivers highly unique slates customized to individual users.",
      "HIGH BREADTH (retrieval_ann_20): Intra-list Diversity is high (0.58), indicating slates feature a wide variety of semantic genres/topics."
    ]
  },
  "embedding_hubness": {
    "ranker_20": [
      "MODERATE HUBNESS (ranker_20): The retrieval distribution is moderately right-skewed (Skewness: 3.7). A handful of items are dominating slates, but they are not strictly tied to the global user centroid (r=0.15)."
    ],
    "retrieval_ann_100": [
      "HEALTHY LATENT SPACE (retrieval_ann_100): Skewness is low (1.4), indicating recommendation volume is naturally distributed without geometric black holes."
    ],
    "retrieval_ann_20": [
      "HEALTHY LATENT SPACE (retrieval_ann_20): Skewness is low (2.8), indicating recommendation volume is naturally distributed without geometric black holes."
    ]
  },
  "executive_summary": [
    "TWO-STAGE FUNNEL SUMMARY (@100 -> @20): Stage-1 ANN Recall@100 is 1.60%. Unranked ANN Recall@20 retains 17.2% of that pool (0.27%), while Stage-2 GATv2 Ranker@20 captures 66.6% of the pool (1.06%), delivering +287.3% Recall@20, +377.3% NDCG@20 (0.0023 -> 0.0108), and +396.3% MRR@20 (0.0055 -> 0.0275) lift over ANN@20.",
    "TIER-STRATIFIED RANKING BREAKDOWN (@20): Composite NDCG@20 shifted from 0.0017 (Retriever) to 0.0055 (Ranker) | Head (Tier 0): 0.0016 -> 0.0114 | Torso (Tier 1): 0.0019 -> 0.0043 | Tail (Tier 2): 0.0018 -> 0.0011.",
    "HIGH SPHERICAL STABILITY (Movie Candidate): Mean Resultant Length R_bar = 0.9997 (P10 = 0.9992, Median vMF kappa = 51349.1), Mean Geodesic Deviation = 1.2 deg (P90 = 1.8

deg, Robust MAD = 0.3 deg). 100.0% of entities have tight centroids (R_bar >= 0.90), while 0.0% exhibit strong context shift (R_bar < 0.70).",
    "HIGH SPHERICAL STABILITY (User Query): Mean Resultant Length R_bar = 0.9996 (P10 = 0.9989, Median vMF kappa = 114995.2), Mean Geodesic Deviation = 0.7 deg (P90 = 1.1 deg, Robust MAD = 0.1 deg). 100.0% of entities have tight centroids (R_bar >= 0.90), while 0.0% exhibit strong context shift (R_bar < 0.70)."
  ],
  "gatv2_spherical_fidelity": [
    "HIGH SPHERICAL STABILITY (Movie Candidate): Mean Resultant Length R_bar = 0.9997 (P10 = 0.9992, Median vMF kappa = 51349.1), Mean Geodesic Deviation = 1.2 deg (P90 = 1.8 deg, Robust MAD = 0.3 deg). 100.0% of entities have tight centroids (R_bar >= 0.90), while 0.0% exhibit strong context shift (R_bar < 0.70).",
    "HIGH SPHERICAL STABILITY (User Query): Mean Resultant Length R_bar = 0.9996 (P10 = 0.9989, Median vMF kappa = 114995.2), Mean Geodesic Deviation = 0.7 deg (P90 = 1.1 deg, Robust MAD = 0.1 deg). 100.0% of entities have tight centroids (R_bar >= 0.90), while 0.0% exhibit strong context shift (R_bar < 0.70)."
  ],
  "popularity_bias": {
    "ranker_20": [
      "NICHE EXPLORATION (ranker_20): The model actively suppresses popularity, surfacing items 42.5% less popular than organic consumption.",
      "POWER USER PENALTY (ranker_20): Power users (Tier 0) experience significantly worse popularity bias than Light users (Tier 2). The model ignores their rich histories and defaults to blockbusters."
    ],
    "retrieval_ann_100": [
      "NICHE EXPLORATION (retrieval_ann_100): The model actively suppresses popularity, surfacing items 73.9% less popular than organic consumption.",
      "POWER USER PENALTY (retrieval_ann_100): Power users (Tier 0) experience significantly worse popularity bias than Light users (Tier 2). The model ignores their rich histories and defaults to blockbusters."
    ],
    "retrieval_ann_20": [
      "NICHE EXPLORATION (retrieval_ann_20): The model actively suppresses popularity, surfacing items 73.7% less popular than organic consumption.",
      "POWER USER PENALTY (retrieval_ann_20): Power users (Tier 0) experience significantly worse popularity bias than Light users (Tier 2). The model ignores their rich histories and defaults to blockbusters."
    ]
  },
  "rank_agreement_and_churn": [
    "MASSIVE CHURN: The Ranker completely discards and replaces over 80% of the Retriever's top items (Overlap: 19.7%). It is acting as an extremely aggressive filter.",
    "AGGRESSIVE RE-RANKING: The Ranker drastically scrambles the relative ordering of retrieved items (Spearman ρ = -0.01), proving it scores candidate items very differently than the ANN space."
  ],
  "stratified_metrics": {
    "ranker_20": [

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    "MODERATE RECALL LIFT: Global Recall@20 (1.1%) beats the random baseline (0.5%), but there is room for improvement.",
    "MODERATE PRECISION LIFT: Global Precision@20 (0.61%) is better than random (0.24%), suggesting basic relevance filtering is functioning.",
    "SLOW SURFACING WARNING: Global MRR@20 is heavily degraded (0.03), indicating users must scroll significantly to find the first relevant item.",
    "EXCELLENT HISTORY UTILIZATION: Power users (Tier 0 Recall: 1.4%) perform massively better than light users (Tier 2 Recall: 0.9%). The model successfully translates rich interaction histories into highly relevant slates.",
    "HEALTHY RANKING STRATIFICATION: NDCG is higher for power users (Tier 0: 2.01% vs Tier 2: 0.66%). The model not only retrieves the right items for power users but places them higher in the slate.",
    "POPULARITY BIAS DETECTED: Ranking quality heavily favors head/popular movies (Tier 0 NDCG: 1.14%) compared to long-tail/niche items (Tier 2 NDCG: 0.11%)."
   ],
   "retrieval_ann_100": [
    "RECALL FAILURE: The model fails to outperform a random candidate generator on Recall.",
    "PRECISION FAILURE: The model retrieves slates with the same or worse precision than a completely random draw.",
    "SLOW SURFACING WARNING: Global MRR@100 is heavily degraded (0.01), indicating users must scroll significantly to find the first relevant item.",
    "EXCELLENT HISTORY UTILIZATION: Power users (Tier 0 Recall: 2.2%) perform massively better than light users (Tier 2 Recall: 1.4%). The model successfully translates rich interaction histories into highly relevant slates.",
    "HEALTHY RANKING STRATIFICATION: NDCG is higher for power users (Tier 0: 1.31% vs Tier 2: 0.35%). The model not only retrieves the right items for power users but places them higher in the slate.",
    "POPULARITY BIAS DETECTED: Ranking quality heavily favors head/popular movies (Tier 0 NDCG: 0.59%) compared to long-tail/niche items (Tier 2 NDCG: 0.38%)."
   ],
   "retrieval_ann_20": [
    "RECALL FAILURE: The model fails to outperform a random candidate generator on Recall.",
    "PRECISION FAILURE: The model retrieves slates with the same or worse precision than a completely random draw.",
    "SLOW SURFACING WARNING: Global MRR@20 is heavily degraded (0.01), indicating users must scroll significantly to find the first relevant item.",
    "EXCELLENT HISTORY UTILIZATION: Power users (Tier 0 Recall: 0.5%) perform massively better than light users (Tier 2 Recall: 0.1%). The model successfully translates rich interaction histories into highly relevant slates.",
    "HEALTHY RANKING STRATIFICATION: NDCG is higher for power users (Tier 0: 0.61% vs Tier 2: 0.05%). The model not only retrieves the right items for power users but places them higher in the slate.",
    "STRONG LONG-TAIL PERFORMANCE: Model successfully surfaces and ranks niche/long-tail items (Tier 2 NDCG: 0.18%) effectively relative to popular head items (Tier 0 NDCG: 0.16%)."
   ]
  }
```

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}
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